



A Case Study of Classroom Podcast in Ohlone College

A
Dissertation
Presented to the
Faculty of the Hufstedler School of Education
Alliant International University

by
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San Francisco, 2010



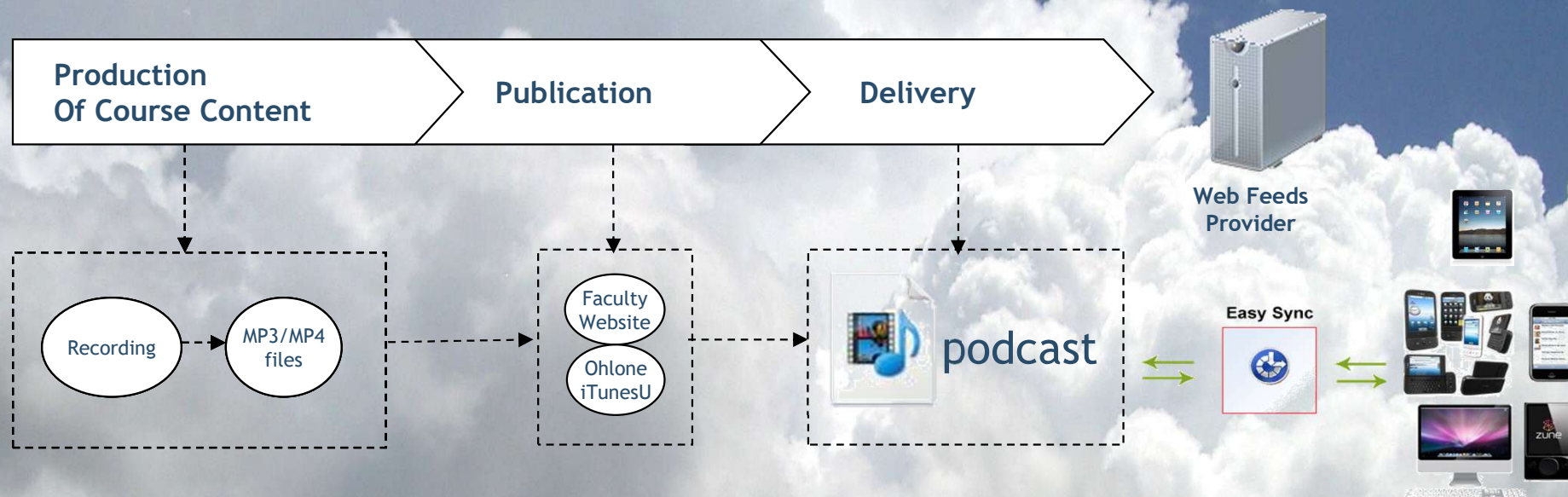


- Since the 1990s Internet and Web technologies have created opportunities for exchange information, technology-assisted teaching and learning, as well as online and distance learning for higher education.
- Like the Internet and Web technologies, emerging mobile and entertainment technologies adds new dimensions for students to learn, communicate, and exchange information.
- According to the Pew Internet and American Life Project (2009) showed that mobile phone use had climbed 63% in the 12- to 17-year-old group from fall 2006 to 71% in early 2008. They also reported that 74% of teens owned an iPod or MP3 player and 60% of teens owned a personal computer.





Ohlone College's Podcasting Service



Internet





Ohlone College's iTunesU



File Edit View Controls Store Advanced Help

iTunes

Ohlone College > History > U.S. History 117A - Prof. Darren L. Bardell - Tab 1

U.S. History 117A - Prof. Darren L. Bardell LOGOUT

Last Modified: Dec 10, 2009
Tracks in Tab 1: 19

GET TRACKS
SUBSCRIBE

Tab 1	Tab 2	Name	Time	Artist	Album	Price
1		The New England Way	1:17:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
2		Mercantilism & Enlightenment	53:15	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
3		Ben Franklin & Stage 1 of Rev	1:19:10	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
4		Ch 6/2 part 1: Road To Revolution 2/26	50:06	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free GET
5		Ch 6/2 part 2: Phases of Revolution 2/28	52:11	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free GET
6		Final Phases to Revolution	1:20:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
7		Revolutionary War	1:13:31	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
8		The Experiment Begins	1:20:29	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
9		President Washington	1:31:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
10		Jefferson's Revolutions	39:36	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
11		Early National Period	1:13:12	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
12		Mainfest Destiny	1:04:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
13		American Exceptionalism	1:25:58	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
14		Economic Development North v South	1:24:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
15		War With Mexico	1:32:46	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
16		Prelude to Civil War	1:26:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
17		Civil War part 1	1:02:40	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
18		Civil War Part 2 + Reconstruction	1:27:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
19		Reconstruction And Final exam Review	1:15:27	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET

19 songs

Podcasting
reinforces
learning



Easy Sync



Ohlone College's iTunesU
<http://itunes.ohlone.edu>





Getting Started with Podcasting

- Digital recording device
- Laptop or PC
- RSS Server (or Web Feed Server)
- Subscribe to RSS Feeds





Statement of the Problem

- Podcasting methods offers new flexible learning outside the classroom, but it also presents challenges to traditional classroom learning.
- One of the major challenges facing classroom learning is how to engage the diverse learning styles of student, especially community college students.
- The benefits of incorporating classroom podcasts into traditional classroom settings remain unclear.





Purpose of the Study

This study examined the use of classroom podcasts as supplemental course materials for playback of lectures and other lessons and the relationship of this use with the test scores.

- (a) The study examined the effects of classroom podcasts and the use of mobile and portable media devices, and the impact of these devices on student learning.
- (b) The prediction was that students who listened to classroom podcasts were more likely to receive higher test scores than those who did not.





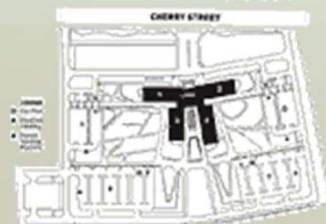
The Site for the Study

- Ohlone College is a part of the Ohlone Community College District, with campuses in Fremont and Newark.
- Ohlone College is located in the proximity of the Silicon Valley, best known for the silicon chip innovators and the headquarters of such leading high-technology manufacturers as Apple Computer, Ebay, Google, Hewlett Packard, Intel, Oracle, Sun, Yahoo, and many more.

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Research Question 1

Are there differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) with regard to the students' test scores?





Research Question 2

Is there a significant difference between the number of times students listened to classroom podcasts before and after the midterm?





Research Question 3

Is there a significant difference between class attendance with regard to the number of hours per week students listened to classroom podcasts?





Research Question 4

Is there a significant difference between the age groups on the number of hours per week students listened to classroom podcasts?





Research Question 5

Is there a significant difference between genders with regard to the number of hours per week students listened to classroom podcasts?





Quasi-Experimental Design

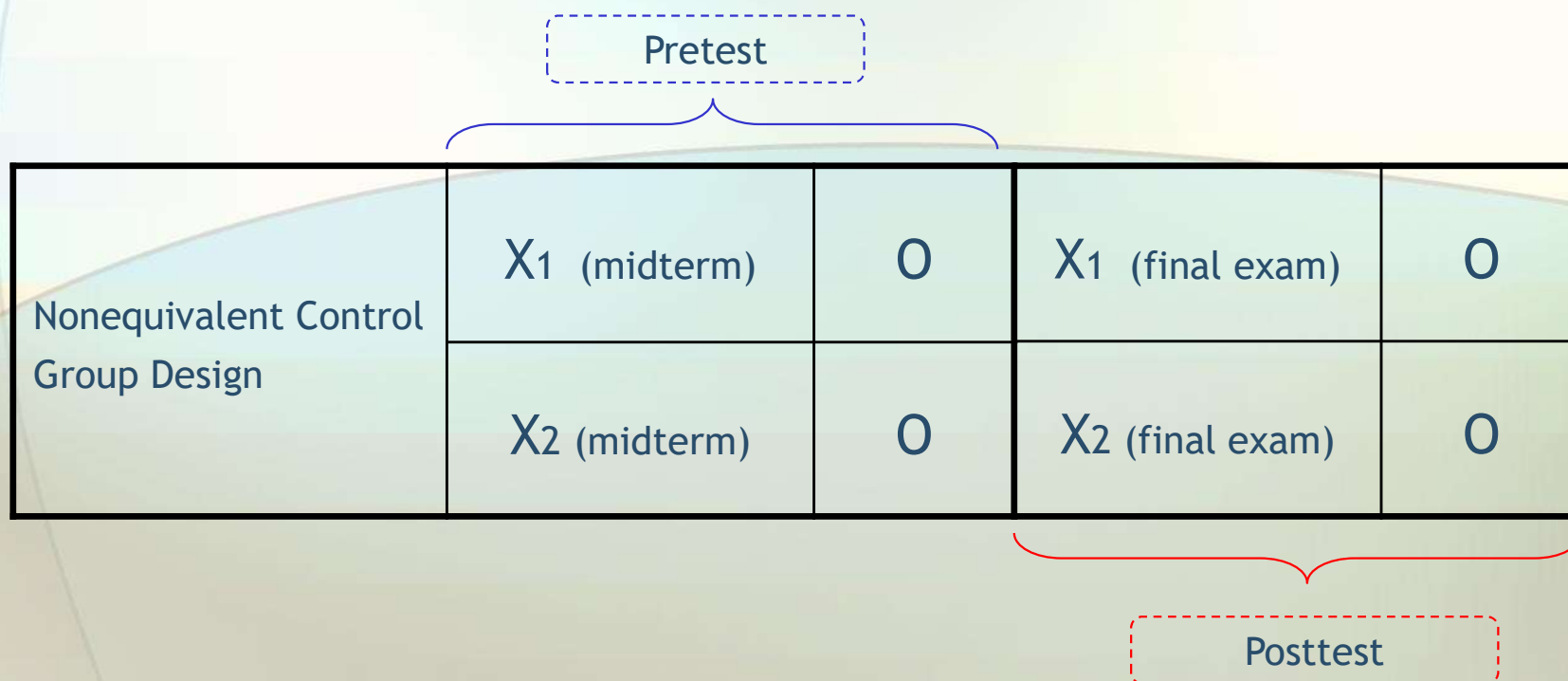


Figure 1: Grouping of participants. X1 = classroom podcasting (experimental treatment), X2 = no classroom podcasting (control treatment), O = posttest.





Participants Grouped by Method of Classroom Instruction

Group	Type of instruction	Designation
1	Traditional classroom instruction	Control group
2	Classroom instruction supplemented with classroom podcast	Treatment group





Treatment Group

- During the Fall 2009 semester, all students from lecture-based course, History of the United States (HIST-117A) were invited to participate in the study.
- All students received traditional classroom instruction supplemented *with* classroom podcasts.
- The study sample of History of the United States for Fall 2009 semester was 72 students (or $N = 72$).





Control Group

- The same lecture-based course from the Spring 2009 semester, taught by the same instructors in traditional classroom instruction without podcasting, was used as a control group.
- All students received traditional classroom instruction with *no* classroom podcast.
- The study sample of History of the United States for the Spring 2009 semester was 70 students (or $N = 70$).





Data Collection Instrument



A three-part handout survey/questionnaire was used to collect students' responses regarding classroom podcast usage:

- (a) demographics questions
- (b) podcasting questions
- (c) open-ended questions





SPSS Data Dictionary



*ClassPodcastsData_Fall09.sav [DataSet1] - SPSS Data Editor

	Name	Type	...	Label	Values	...	A...	Measure
2	InstructionalType	Numeric	2 0	Instructional type	{1, 117A - Classroom Podcasting...	...		Nominal
3	Gender	Numeric	2 0	Gender	{1, Female}...	...		Nominal
4	Major	Numeric	2 0	Academic major	{0, Art}...	...		Nominal
5	Ethnic	Numeric	2 0	Ethnicity	{1, Asian}...	...		Nominal
6	Familiarity	Numeric	2 0	Familiarity with podcasting	{1, Yes}...	...		Nominal
7	Device	Numeric	8 0	Device	{1, MP3 player or Apple iPod}...	...		Nominal
8	Year	Numeric	2 0	Year in college	{1, First}...	...		Ordinal
9	AgeGroup	Numeric	2 0	Age group	{1, 19 years and younger}...	...		Ordinal
10	Attendance	Numeric	2 0	Class attendance	{1, Less than other classes}...	...		Ordinal
11	Access	Numeric	2 0	Perceived ease of access to classroom podcasts	{1, Strongly disagree}...	...		Ordinal
12	Motivation	Numeric	2 0	Motivation levels after the midterm	{1, Strongly disagree}...	...		Ordinal
13	Usefulness	Numeric	2 0	Perceived usefulness of classroom podcasts	{1, Strongly disagree}...	...		Ordinal
14	MidtermScore	Numeric	3 0	Midterm test scores	None	...		Scale
15	FinalScore	Numeric	3 0	Final exam scores	None	...		Scale
16	PretestC	Numeric	3 0	Control group - Pretest	None	...		Scale
17	PosttestC	Numeric	3 0	Control group - Posttest	None	...		Scale
18	PretestT	Numeric	3 0	Treatment group - Pretest	None	...		Scale
19	PosttestT	Numeric	3 0	Treatment group - Posttest	None	...		Scale
20	AverageCourseScore	Numeric	3 2	Average Course test scores	None	...		Scale
21	CourseGrade	Numeric	2 0	Course grade	{1, A}...	...		Scale
22	Frequency	Numeric	2 0	Frequency of use	{0, Zero}...	...		Scale
23	BeforeMidterm	Numeric	2 0	The number of times listen to podcast before the midterm	{0, Zero}...	...		Scale
24	AfterMidterm	Numeric	2 0	The number of times listen to podcast after the midterm	{0, Zero}...	...		Scale
25	SamePodcast	Numeric	2 0	Number of times listen to the same podcast	{0, Zero}...	...		Scale
26	Quality	String	...	The quality of classroom podcasts	None	...		Nominal
27	LearningTool	String	...	Podcasting as a learning tool	None	...		Nominal
28	Preference	String	...	Learning preferences	None	...		Nominal
29	Improvement	String	...	Improvements	None	...		Nominal

Data View Variable View

SPSS Processor is ready





Demographic Profile of Respondents

Gender	<i>n</i>	%
Female	32	52.5
Male	29	47.5
Other (self-describe)	-	-

Of the 72 participants, 85% ($n = 61$) responded to the in-class handout survey.





Demographic Profile of Respondents

Age Group	<i>n</i>	%
19-or-younger	31	50.8
20-24 years	25	41
25-29 years	3	4.9
30-or-older	2	3.2





Demographic Profile of Respondents

Ethnicity	<i>n</i>	%
White, Non-Hispanic	26	42.6
Asian	12	19.7
Hispanic	7	11.5
Multi-racial	7	11.5
Filipino	3	4.9
Middle Eastern	3	4.9
Hawaiian	2	3.3
Greek	1	1.6





Demographic Profile of Respondents

Year in college	<i>n</i>	%
First year	8	13.1
Second year	36	59.0
Third year	9	14.8
Fourth year	5	8.2
Fifth year or more	3	4.9





Demographic Profile of Respondents

Academic major	<i>n</i>	%
Art	12	20.0
Business	10	23.3
Communication	4	16.7
Engineering	4	6.7
Science	14	15.0
Social Science	9	8.3
Undecided	5	6.7
Other	2	3.3





Demographic Profile of Respondents

Class attendance	<i>n</i>	%
Less than other classes	2	3.3
Same as other classes	42	70.0
More than other classes	16	26.7





Demographic Profile of Respondents

Listening device	<i>n</i>	%
MP3/iPod player	10	16.7
Personal computer (PC)	32	53.3
MP3 player and PC	13	21.7
Cell phone/PDA, MP3, and PC	4	6.7
Cell phone/PDA and PC	1	1.7





Students' Attitudes Toward the Use of Classroom Podcasts

Response	Perceived ease of use (%)	Perceived usefulness (%)
Strongly Agree	32.8	18.0
Agree	44.3	45.9
Neutral	23.0	26.2
Disagree	0.0	4.9
Strongly Disagree	0.0	3.3





Results of Research Question 1

- An independent-samples t test was calculated for the midterm (pretest) test scores of the treatment group (pretest $n = 72$) and comparison group (pretest $n = 70$).

X_1 (midterm)	0	X_1 (final exam)	0
X_2 (midterm)	0	X_2 (final exam)	0

- The results indicated ***no*** statistically significant differences between the two groups ($t = 1.32$, $df = 126.77$, $p < .19$).





Results of Research Question 1

- A repeated independent-samples t test was calculated for the final examination (posttest) of the treatment group (posttest $n = 72$) and control group (posttest $n = 70$).

X_1 (midterm)	0	X_1 (final exam)	0
X_2 (midterm)	0	X_2 (final exam)	0

- The results indicated ***no*** statistically significant differences between the two groups ($t = .30$, $df = 129.74$, $p < .765$).





Results of Research Question 1

- A paired-samples t test was calculated for repeated measures (pretest and posttest) on the treatment group.

X_1 (midterm)	0	X_1 (final exam)	0
X_2 (midterm)	0	X_2 (final exam)	0

- The results indicated that the difference was statistically significant ($N = 72$; $df = 71$; $p < .038$).





Results for Research Question 2

- A 5-point Likert-type scale (0 = *not at all*, 1 = *once*, 2 = *twice*, 3 = *three times*, and 4 = *four or more times*) was used.
- Analysis of the 60 matched pairs by a paired-samples *t* test showed a significant difference between the number of times students listened to classroom podcasts before the midterm ($M = 1.33$, $SD = 1.49$) and after the midterm ($M = 1.75$, $SD = 1.32$).
- The difference was statistically significant ($N = 60$, $df = 59$, $p < .012$).





Results for Research Question 3

- Three levels class attendance (1 = *less than other classes*, 2 = *same as other classes*, 3 = *more than other classes*)
- The number of hours per week students listened to classroom podcasts (0 = *not at all*, 1 = *1 hour or less*, 2 = *2 hours*, and 3 = *3 hours or more*)
- A one-way ANOVA was used to determine statistically significant difference between class attendance and number of hours the students listened to classroom podcasts.
- The difference was *no* statistically significant, $F(2, 57) = .132$, $p < .877$.





Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

Class attendance	<i>n</i>	<i>M</i>	<i>SD</i>
Less than other classes	2	1.00	0.000
Same as other classes	42	1.21	0.925
More than other classes	16	1.31	0.946
Total	60	1.23	0.909





A One-Way ANOVA of Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

	<i>df</i>	<i>F</i>	Significance (2-tailed)
Frequency of use	2	1.32	.877





Results for Research Question 4

- Three levels age groups (1 = 19 or younger, 2 = 20-24, 3 = 25 or older)
- The number of hours per week students listened to classroom podcasts (0 = *not at all*, 1 = 1 hour or less, 2 = 2 hours, and 3 = 3 hours or more)
- A one-way ANOVA was used to determine statistical significance of the differences among age groups according to the number of hours per week they listened to classroom podcasts.
- The difference was *no* statistically significant, $F(2,57) = 2.01, p < .144$.





Age Group for the Number of Hours Per Week Students Listened to Classroom Podcasts

Age group	<i>n</i>	<i>M</i>	<i>SD</i>
19 years or younger	30	1.17	.874
20 to 24 years	25	1.16	.943
25 years or older	5	2	.707
Total	60	1.23	.909





A One-Way ANOVA of Age Groups and Number of Hours Per Week Students Listened to Classroom Podcasts

	<i>df</i>	<i>F</i>	Significance (2-tailed)
Frequency of use	2	2.01	.144





Results for Research Question 5

- Gender (1 = *female* and 2 = *male*)
- The number of hours per week students listened to classroom podcasts (0 = *not at all*, 1 = *1 hour or less*, 2 = *2 hours*, and 3 = *3 hours or more*)
- An independent-samples *t* test was used to compare number of hours per week students listened to classroom podcasts by gender.
- The difference was statistically significant ($df = 60$, $t = 2.54$, $p < .014$).





Number of Hours Per Week Students Listened to Classroom Podcasts by Gender

Gender	<i>n</i>	<i>M</i>	<i>SD</i>
Female	32	1.50	0.916
Male	28	0.93	0.813





An Independent-Samples Two-Tailed t Test for Number of Hours Per Week Students Listened to Classroom Podcasts by Gender

	<i>t</i>	<i>df</i>	Significance (2-tailed)
Frequency of use	2.54	58	.014





Additional Findings



How could podcasts be improved to better meet your needs?

Written responses were assigned to four categories:

- (a) quality of classroom podcasts,
- (b) podcasting as a teaching and learning tool,
- (c) learning preferences, and
- (d) recommendations for improvement.





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Quality of classroom podcasts	<i>f</i>	%
Good the way it is	15	30
Classroom podcasts should be made available to the world	1	2
Did well on the midterm so don't need the podcast	1	2
Technological step in the right direction for students	1	2





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Podcasting as a teaching and learning tool	<i>f</i>	%
Helpful for reviewing class lectures	13	26
Helpful for reviewing interaction between teacher and students	1	2





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Learning preferences	<i>f</i>	%
Preferred classroom lectures	10	20
Preferred audio/video podcast	6	12
Preferred MP3 or compatible audio file format	3	6





Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N =50)

Recommendations for improvement	<i>f</i>	%
Arrange the podcasts in chronological order	4	8
Provide video podcast for visual learners	3	6
Provide better sound quality	3	6
Remove extraneous material from the podcasts	3	6
Expand use of the podcasts to other classes	2	4





Distribution of Final Course Grades for the two groups

Final course grade (Treatment Group)	<i>n</i>	%
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A	22	30.6
B	28	38.9
C	17	23.6
D	4	5.6
F	1	1.4

Final course grade (Control Group)	<i>n</i>	%
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A	24	34.3
B	18	25.7
C	21	30.0
D	4	5.7
F	3	4.3





Instructor Feedback



Positive aspects include:

- (a) students can re-listen to the lectures as often as they like,
- (b) students can listen at their leisure, and
- (c) students who are struggling with learning note-taking skills and English as Second Language (ESL) students can benefit by podcast use.

Disadvantages include:

- (a) students may rely on podcasting and not attend lectures,
- (b) students download classroom podcasts but do not actually listen to them.





Recommendations



- The recommendation for school leaders is to invest in state-of-the-art technology infrastructure and faculty and staff training.
- The recommendation for the people in the leadership positions is to invest in faculty and staff educational training.
- The recommendation for educators should consider technology-based teaching as the leader in student learning.





Suggestion for Future Research



- Replicate this study in other subject areas to test the overall effectiveness of classroom podcasts and determine longitudinal effects and understand students' attitudes toward podcasting.
- On the recommendation of the students, video podcasts is also needed for visual learners. Further study is needed to determine the impact of audio and video podcasts on visual learners.
- A book like device, the iPad (January, 28, 2010), which has the ability to serve for Web surfing, e-reader, game device, audio/video player, and multipurpose multimedia device.





Conclusions



1. This study shows that podcasting has a potential to improve student learning, but it is not for every student.
2. The findings from this study can provide leadership and support for technology-based instruction in higher education.
3. The findings from this study may help educators to determine the necessary balance between podcast learning and traditional classroom interaction.
4. The findings from this study could be used to expand technology-based learning and provide assistance to educators who desire to develop and share podcasts containing course materials with students.
5. Data gathered in this study could be used to provide leadership and support for future directions of technology-based instruction in higher education.





Q & A



Thank you for making a positive
contribution to my education.

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